

CSEC Mathematics
June 2012 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Calculate the EXACT value of

$$\frac{3\frac{1}{5} - \frac{2}{3}}{2\frac{4}{5}}$$

giving your answer as a fraction in its lowest terms.

[3]

$$\text{Numerator} = 3\frac{1}{5} - \frac{2}{3}$$

$$= \frac{16}{5} - \frac{2}{3}$$

$$= \frac{48-10}{15}$$

$$= \frac{38}{15}$$

$$\text{Denominator} = 2\frac{4}{5}$$

$$= \frac{14}{5}$$

$$\text{Numerator} \div \text{Denominator} = \frac{38}{15} \div \frac{14}{5}$$

$$= \frac{38}{15} \times \frac{5}{14}$$

$$= \frac{19}{21}$$

$$\therefore \frac{3\frac{1}{5} - \frac{2}{3}}{2\frac{4}{5}} = \frac{19}{21}$$

(b) The table below shows the cost price, selling price and profit or loss as a percentage of the cost price.

Copy and complete the table below, inserting the missing values at (i) and (ii). [4]

	Cost Price	Selling Price	Percentage Profit or Loss
(i)	\$55.00	\$44.00	20% loss
(ii)	\$80.00	\$100.00	25% profit

(i) Selling Price = \$44.00

Cost Price = \$55.00

Since the selling price is less than the cost price, a loss has occurred.

Loss = Cost Price – Selling Price

= \$55.00 – \$44.00

= \$11.00

$$\begin{aligned}\text{Percentage Loss} &= \frac{\text{Loss}}{\text{Cost Price}} \times 100 \\ &= \frac{11}{55} \times 100 \\ &= 20\%\end{aligned}$$

∴ The percentage loss is 20%.

(ii) A 25% profit implies that the selling price is 125% of the cost price.

$$125\% = \$100$$

$$1\% = \frac{100}{125}$$

$$\begin{aligned}100\% &= \frac{100}{125} \times 100 \\ &= \$80\end{aligned}$$

∴ The cost price is \$80.

(c) The table below shows some rates of exchange:

US \$1.00 = EC \$2.70
TT \$1.00 = EC \$0.40

Calculate the value of

(i) EC \$1 in TT\$ [1]

$$\text{EC } \$0.40 = \text{TT } \$1.00$$

$$\begin{aligned}\text{EC } \$1 &= \frac{\text{TT } \$1.00}{0.40} \\ &= \text{TT } \$2.50\end{aligned}$$

$$\therefore \text{EC } \$1 = \text{TT } \$2.00$$

(ii) US \$80 in EC\$ [1]

$$\text{US } \$1.00 = \text{EC } \$2.70$$

$$\begin{aligned}\text{US } \$80 &= \text{EC } \$2.70 \times 80 \\ &= \text{EC } \$216\end{aligned}$$

$$\therefore \text{US } \$80 = \text{EC } \$216$$

(iii) TT \$648 in US\$

$$\text{TT } \$1.00 = \text{EC } \$0.40$$

$$\text{TT } \$648 = \text{EC } \$0.40 \times 648$$

$$= \text{EC } \$259.20$$

Now,

$$\text{US } \$1.00 = \text{EC } \$2.70$$

$$\text{EC } \$2.70 = \text{US } \$1.00$$

$$\text{EC } \$1.00 = \frac{\text{US } \$1.00}{2.70}$$

$$\text{EC } \$259.20 = \frac{\text{US } \$1.00}{2.70} \times 259.20$$

$$= \text{US } \$96.00$$

$$\therefore \text{TT } \$648 = \text{US } \$96.00$$

Total: 11 marks

2. (a) Factorise completely:

(i) $2x^3y + 6x^2y^2$ [2]

$$2x^3y + 6x^2y^2 = 2x^2y(x + 3y)$$

(ii) $9x^2 - 4$ [1]

$$9x^2 - 4 = (3x - 2)(3x + 2) \quad [\text{difference of two squares}]$$

(iii) $4x^2 + 8xy - xy - 2y^2$ [2]

$$\begin{aligned} &4x^2 + 8xy - xy - 2y^2 \\ &= 4x(x + 2y) - y(x + 2y) \\ &= (x + 2y)(4x - y) \end{aligned}$$

(b) Solve for x :

$$\frac{2x-3}{3} + \frac{5-x}{2} = 3 \quad [3]$$

$$\frac{2x-3}{3} + \frac{5-x}{2} = 3$$

$$\frac{2(2x-3)+3(5-x)}{6} = 3$$

$$\frac{4x-6+15-3x}{6} = 3$$

$$\frac{x+9}{6} = 3$$

$$x + 9 = 6 \times 3$$

$$x + 9 = 18$$

$$x = 18 - 9$$

$$x = 9$$

(c) Solve the simultaneous equations:

$$3x - 2y = 10$$

$$2x + 5y = 13$$

[4]

$$3x - 2y = 10 \quad \dots \text{Equation 1}$$

$$2x + 5y = 13 \quad \dots \text{Equation 2}$$

Multiplying Equation 1 by 2 gives:

$$6x - 4y = 20 \quad \dots \text{Equation 3}$$

Multiplying Equation 2 by 3 gives:

$$6x + 15y = 39 \quad \dots \text{Equation 4}$$

So, we have,

$$6x - 4y = 20 \quad \dots \text{Equation 3}$$

$$6x + 15y = 39 \quad \dots \text{Equation 4}$$

Equation 3 – Equation 4 gives,

$$-19y = -19$$

$$y = \frac{-19}{19}$$

$$y = 1$$

Now,

Substituting $y = 1$ into Equation 1 gives,

$$3x - 2(1) = 10$$

$$3x - 2 = 10$$

$$3x = 10 + 2$$

$$3x = 12$$

$$x = \frac{12}{3}$$

$$x = 4$$

$$\therefore x = 4 \text{ and } y = 1.$$

Total: 12 marks

3. (a) In a survey of 36 students, it was found that

30 play tennis,

x play volleyball ONLY,

$9x$ play BOTH tennis and volleyball,

4 play neither tennis nor volleyball.

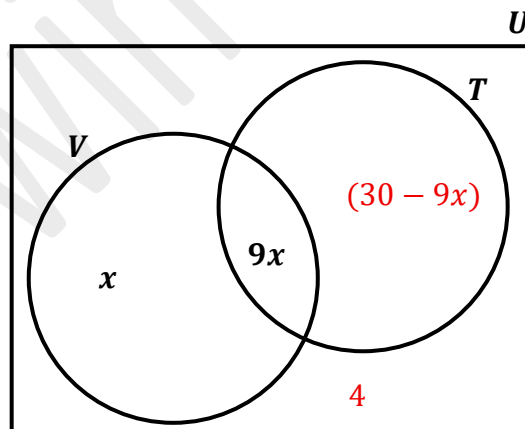
(i) Given that:

$U = \{\text{students in the survey}\}$

$V = \{\text{students who play Volleyball}\}$

$T = \{\text{students who play Tennis}\}$

Copy and complete the Venn diagram below to show the number of students in the subsets marked y and z . [2]



$$y = 30 - 9x$$

$$z = 4$$

- (ii) (a) Write an **expression** in x to represent the TOTAL number of students in the survey. [1]

The number of students who play volleyball only = x

The number of students who play volleyball only = $30 - 9x$

The number of students who play both tennis and volleyball = $9x$

The number of students who do not play either tennis or volleyball = 4

Therefore,

$$\begin{aligned}\text{Number of students in the survey} &= x + (30 - 9x) + 9x + 4 \\ &= x + 30 - 9x + 9x + 4 \\ &= x + 34\end{aligned}$$

- (b) Write an **equation** in x to represent the total number of students in the survey and hence solve for x . [2]

The total number of students = 36

The expression for the total number of students = $x + 34$

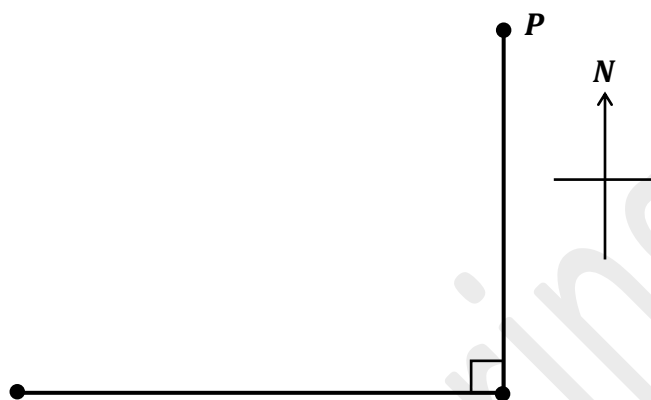
Hence, the equation is

$$x + 34 = 36$$

$$x = 36 - 34$$

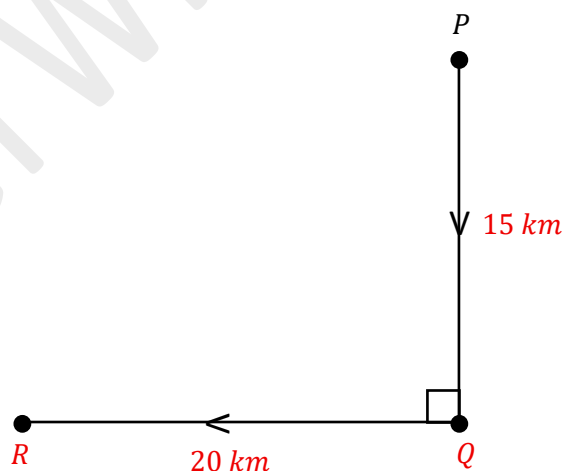
$$x = 2$$

- (b) The diagram below, not drawn to scale, shows the journey of a ship which started at port P , sailed 15 km due south to port Q , and then a further 20 km due west to port R .



- (i) Copy the diagram and label it to show the points Q and R , and the distances 20 km and 15 km .

[2]



- (ii) Calculate PR , the **shortest** distance of the ship from the port where the journey started. [2]

Using Pythagoras' Theorem,

$$PR^2 = PQ^2 + RQ^2$$

$$PR^2 = (15)^2 + (20)^2$$

$$PR^2 = 225 + 400$$

$$PR^2 = 625$$

$$PR = \sqrt{625}$$

$$PR = 25 \text{ km}$$

- (iii) Calculate the measure of angle QPR , giving your answer to the nearest degree. [3]

$$\begin{aligned} \sin Q\hat{P}R &= \frac{\text{opp}}{\text{hyp}} \\ &= \frac{20}{25} \end{aligned}$$

$$\therefore Q\hat{P}R = \sin^{-1}\left(\frac{20}{25}\right)$$

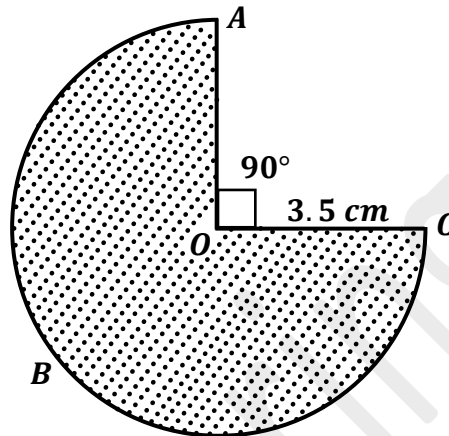
$$= 53.1^\circ$$

$$= 53^\circ \quad (\text{to the nearest degree})$$

Total: 12 marks

4. The diagram below, **not drawn to scale**, shows the cross section of a prism in the shape of a sector of a circle, centre O , and radius 3.5 cm. The angle at the centre is 270° .

Use $\pi = \frac{22}{7}$



(a) Calculate

- (i) the length of the arc ABC

[2]

$$\begin{aligned}\text{Length of the arc } ABC &= \frac{270^\circ}{360^\circ} \times 2\pi r \\ &= \frac{3}{4} \times 2 \times \frac{22}{7} \times 3.5 \\ &= 16.5 \text{ cm}\end{aligned}$$

- (ii) the perimeter of the sector $OABC$

[2]

$$\begin{aligned}\text{Perimeter of the sector } OABC \\ &= \text{Length of } AO + \text{Arc length } ABC + \text{Length of radius } CO\end{aligned}$$

$$= 16.5 + 3.5 + 3.5$$

$$= 23.5 \text{ cm}$$

(iii) the area of the sector $OABC$

[2]

$$\begin{aligned} \text{Area of the sector } OABC &= \frac{270^\circ}{360^\circ} \times \pi r^2 \\ &= \frac{3}{4} \times \frac{22}{7} \times (3.5)^2 \\ &= 28.875 \text{ cm}^2 \end{aligned}$$

$\therefore$ The area of the sector $OABC$ is 28.875 cm^2 .

(b) The prism is 20 cm long and is a solid made of tin. Calculate

(i) the volume of the prism

[2]

$$\begin{aligned} \text{Volume of the prism} &= \text{Cross-sectional Area} \times \text{Height} \\ &= 28.875 \times 20 \\ &= 577.5 \text{ cm}^3 \end{aligned}$$

$\therefore$ The volume of the prism is 577.5 cm^3 .

- (ii) the mass of the prism, **to the nearest kg** , given that 1 cm^3 of tin has a mass of 7.3 kg . [2]

$$1\text{ cm}^3 \text{ of tin weighs } = 7.3\text{ g}$$

$$\begin{aligned} 577.5\text{ cm}^3 \text{ of tin weighs } &= 7.3 \times 577.5 \\ &= 4\,215.75\text{ g} \end{aligned}$$

Now,

$$1000\text{ g} = 1\text{ kg}$$

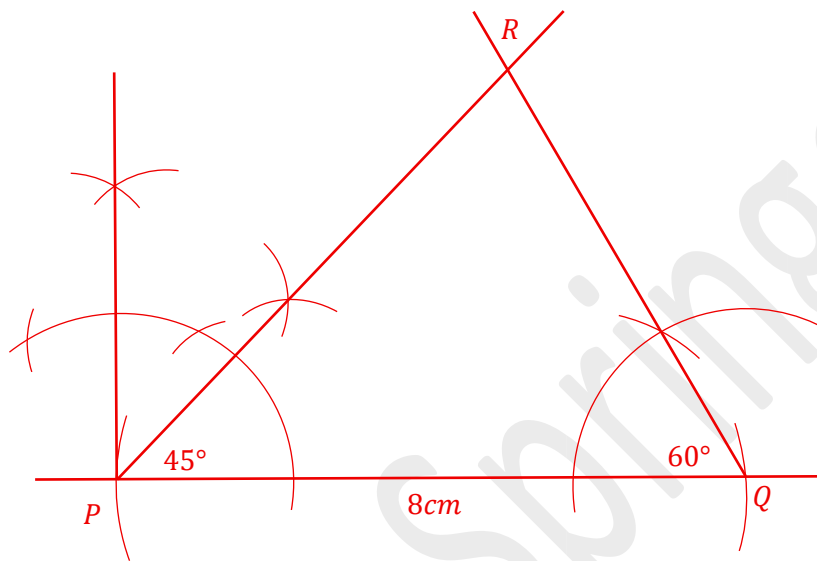
$$1\text{ g} = \frac{1}{1000}\text{ kg}$$

$$\begin{aligned} 4\,215.75\text{ g} &= \frac{1}{1000} \times 4\,215.75 \\ &= 4.21575\text{ kg} \\ &= 4.2\text{ kg} \quad (\text{to the nearest } kg) \end{aligned}$$

$\therefore$ The mass of the prism is 4.2 kg .

Total: 10 marks

5. (a) (i) Using a ruler, a pencil and a pair of compasses, construct triangle PQR with $PQ = 8\text{ cm}$, $\angle PQR = 60^\circ$ and $\angle QPR = 45^\circ$. [4]



- (ii) Measure and state the length of RQ . [1]

By measurement using a ruler, the length of $RQ = 6\text{ cm}$.

- (b) The line l passes through the points $S(6, 6)$ and $T(0, -2)$.

Determine

- (i) the gradient of the line, l [2]

The points are $S(6, 6)$ and $T(0, -2)$.

$$\begin{aligned}
 \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{6 - (-2)}{6 - 0} \\
 &= \frac{8}{6} \\
 &= \frac{4}{3}
 \end{aligned}$$

(ii) the equation of the line, l [2]

Since $(0, -2)$ lies on the line, this implies that $c = -2$.

In part (b)(i), we found that $m = \frac{4}{3}$.

Therefore, the equation of the line is $y = \frac{4}{3}x - 2$.

(iii) the midpoint of the line segment, TS [1]

Point $T(0, -2)$ and Point $S(6, 6)$.

$$\begin{aligned}
 \text{Midpoint} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\
 &= \left(\frac{6 + 0}{2}, \frac{6 + (-2)}{2} \right) \\
 &= \left(\frac{6}{2}, \frac{4}{2} \right) \\
 &= (3, 2)
 \end{aligned}$$

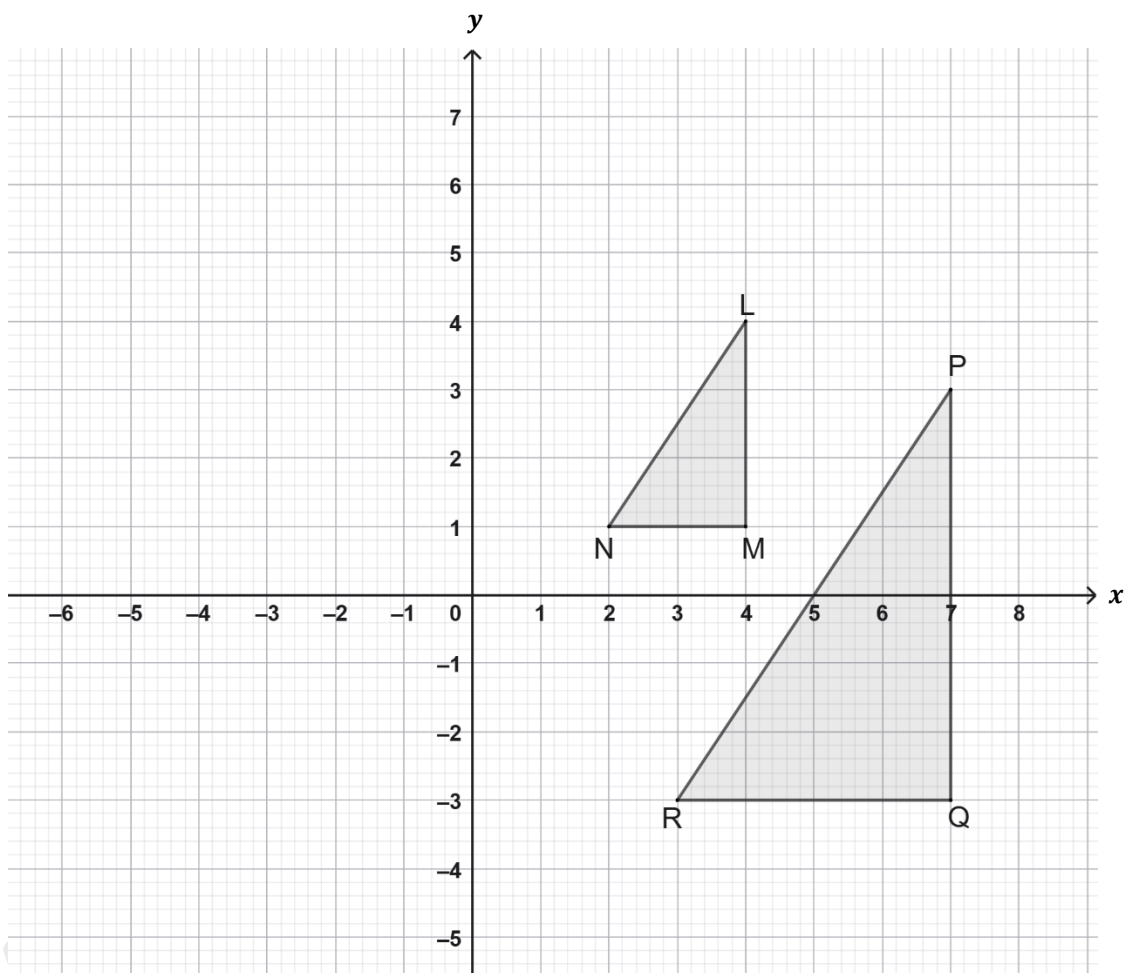
(iv) the length of the line segment, TS .

$$\begin{aligned}\text{Length of line segment } TS &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(6 - 0)^2 + (6 - (-2))^2} \\ &= \sqrt{6^2 + 8^2} \\ &= \sqrt{36 + 64} \\ &= \sqrt{100} \\ &= 10 \text{ units}\end{aligned}$$

Total: 12 marks

6. An answer sheet is provided for this question.

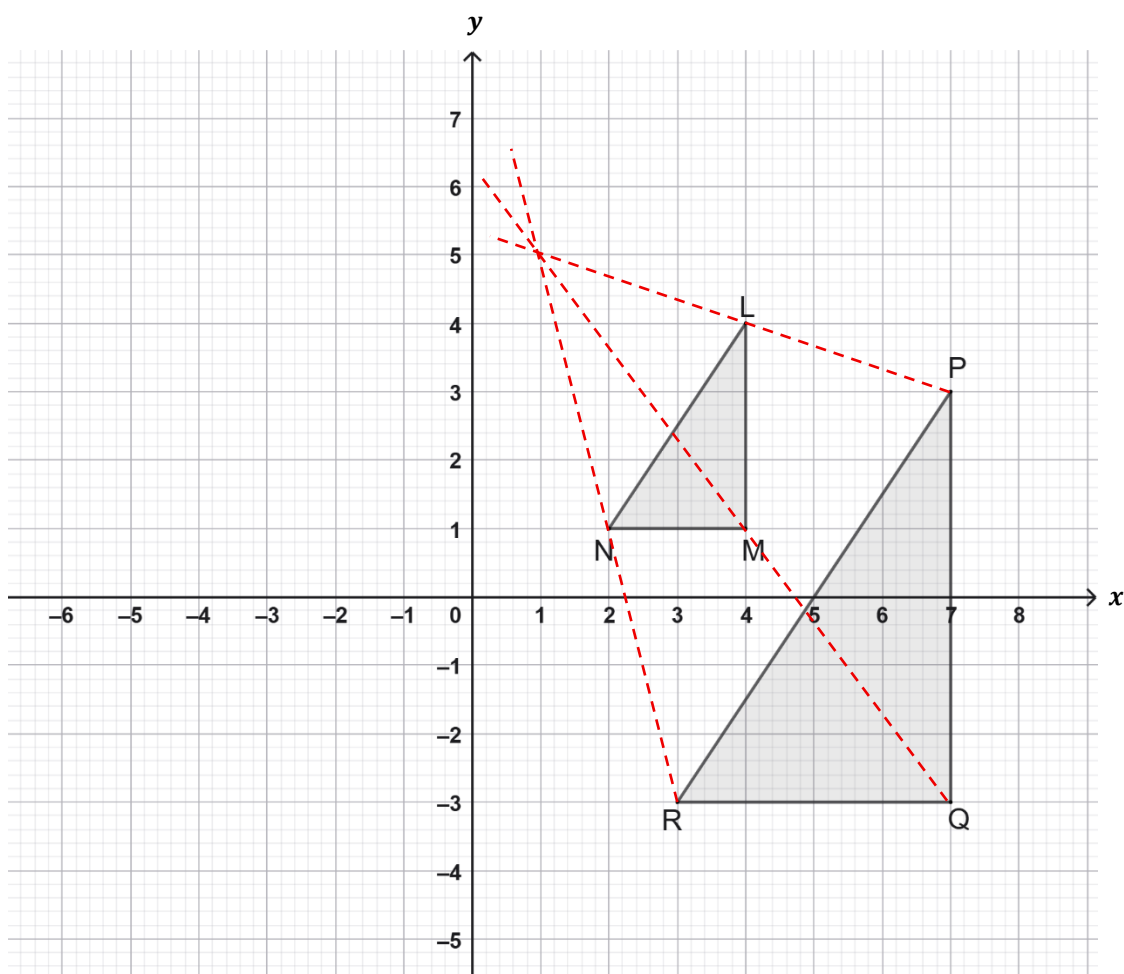
The graph below shows triangle LMN and its image PQR after an enlargement.



On the answer sheet provided,

- (a) Locate the centre of enlargement, showing your method clearly.

[2]



∴ The centre of enlargement is (1, 5).

(b) State the scale factor and the coordinates of the centre of the enlargement.

[2]

$$\begin{aligned}\text{Scale factor} &= \frac{\text{Length of } PQ}{\text{Length of } LM} \\ &= \frac{6}{3} \\ &= 2\end{aligned}$$

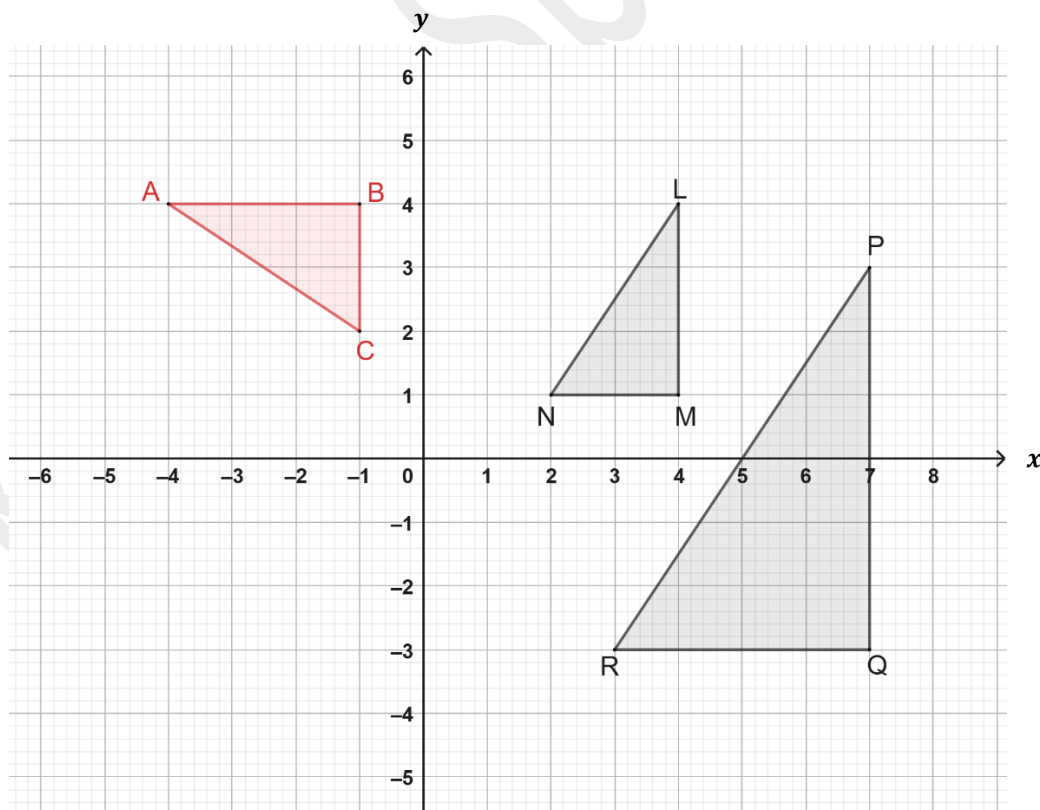
The coordinates of the centre of enlargement is (1, 5).

(c) Determine the value of $\frac{\text{Area of } PQR}{\text{Area of } LMN}$. [2]

Since $\frac{RQ}{NM} = \frac{2}{1}$, then

$$\begin{aligned}\frac{\text{Area of } PQR}{\text{Area of } LMN} &= \left(\frac{2}{1}\right)^2 \\ &= (2)^2 \\ &= 4\end{aligned}$$

(d) Draw and label triangle ABC with coordinates $(-4, 4)$, $(-1, 4)$ and $(-1, 2)$ respectively. [2]



- (e) Describe fully the single transformation which maps triangle LMN onto triangle ABC . [3]

The transformation which maps $\triangle LMN$ into $\triangle ABC$ is a rotation of 90° in an anti-clockwise direction, about the origin, $(0,0)$.

Total: 11 marks

7. The table below shows the ages, to the nearest year, of the persons who visited the clinic during a particular week.

Age (yrs)	Number of persons	Cumulative Frequency
40 – 49	4	4
50 – 59	11	15
60 – 69	20	
70 – 79	12	
80 – 89	3	50

- (a) Copy and complete the table to show the cumulative frequency. [2]

The completed table is shown below:

Age (yrs)	Number of persons	Cumulative Frequency
40 – 49	4	4
50 – 59	11	15
60 – 69	20	35
70 – 79	12	47
80 – 89	3	50

(b) Using a scale of 2 *cm* to represent 10 years on the *x*-axis and 1 *cm* to represent 5 persons on the *y*-axis, draw the cumulative frequency curve for the data.

[5]

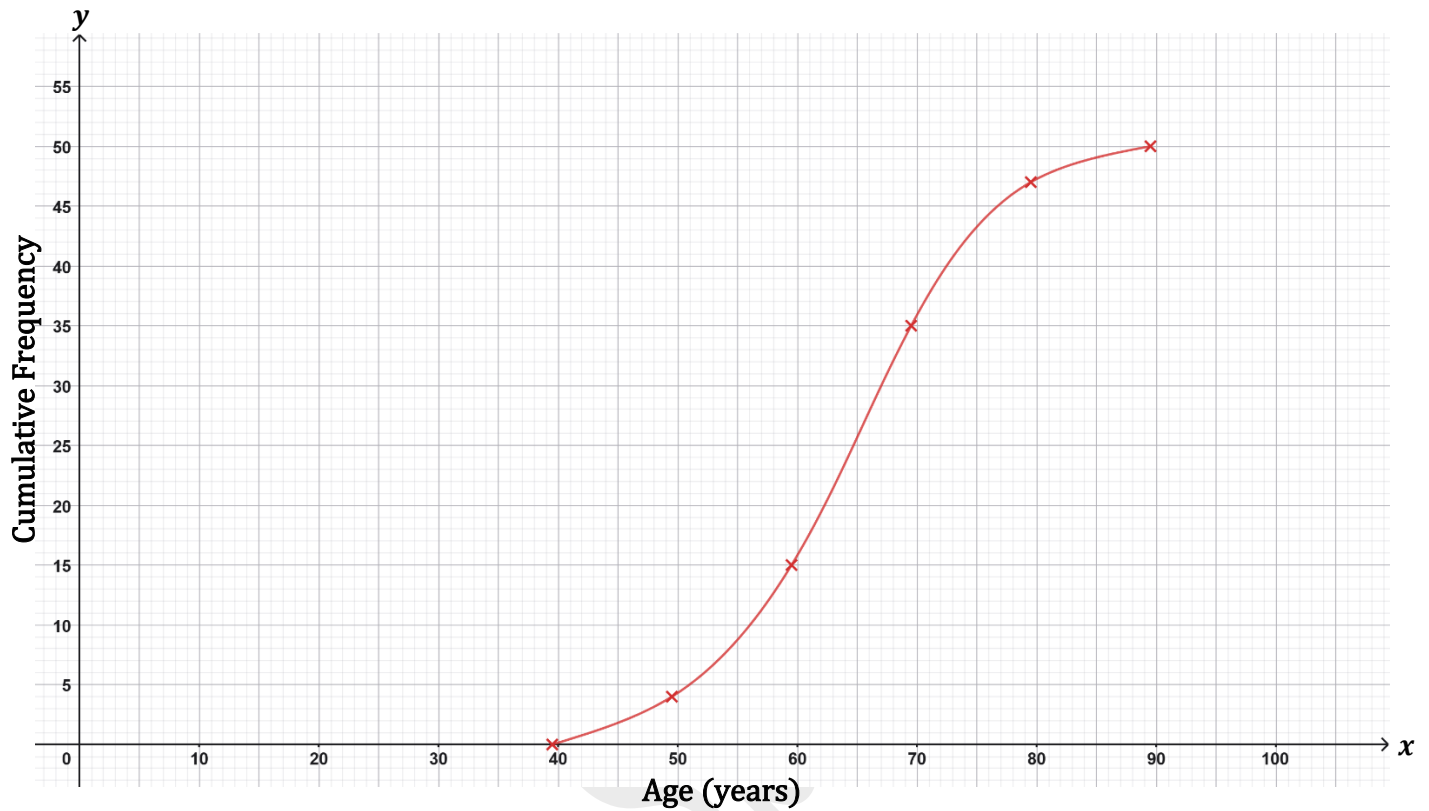
Age (yrs)	Number of persons	Cumulative Frequency	Points to plot (U.C.B. , Cumulative Frequency)
			(39.5, 0)
40 – 49	4	4	(49.5, 4)
50 – 59	11	15	(59.5, 15)
60 – 69	20	35	(69.5, 35)
70 – 79	12	47	(79.5, 47)
80 – 89	3	50	(89.5, 50)

Scale Used:

x-axis: 2cm represents 10 years

y-axis: 1cm represents 5 persons

The cumulative frequency graph for the data is shown on the next page:

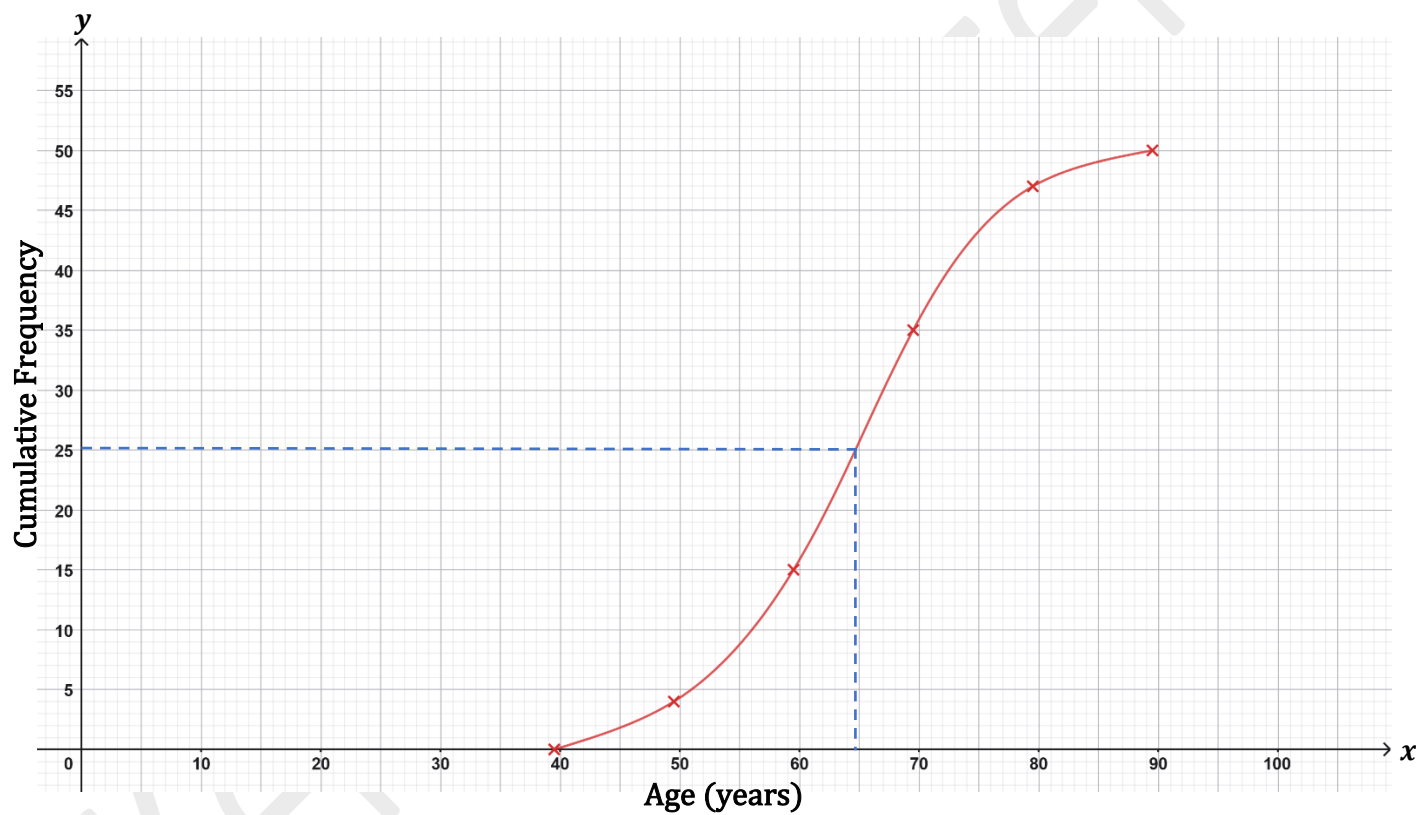


(c) Use your graph drawn at (b) above to estimate

(i) the median age for the data

[2]

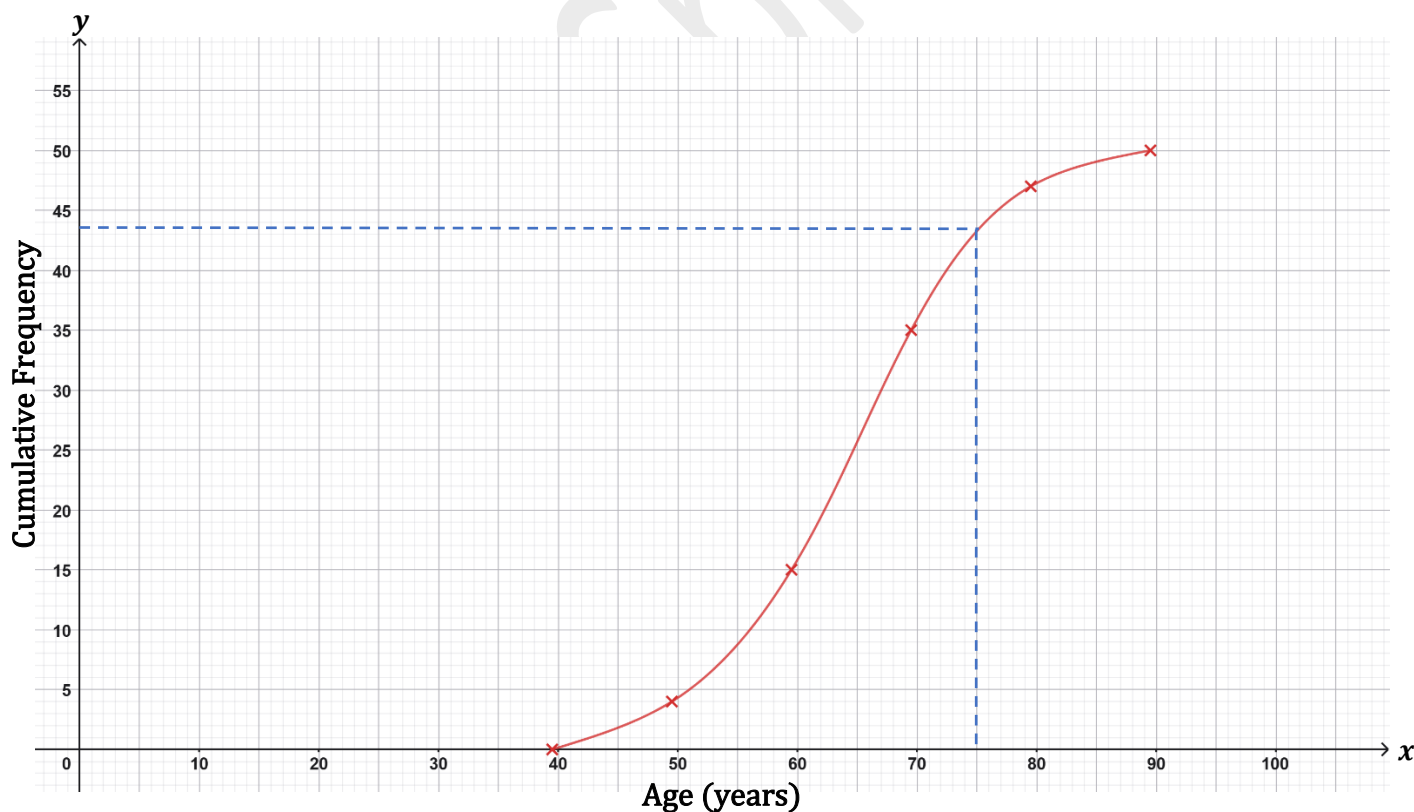
The median age for the data is 64.5 years.



- (ii) the probability that a person who visited the clinic was 75 years or younger. [2]

Draw lines on your graph to show how these estimates were obtained.

$$\begin{aligned} P(\text{person is 75 years or younger}) &= \frac{\text{Number of persons 75 years or younger}}{\text{Total number of persons}} \\ &= \frac{43.5}{50} \\ &= 0.87 \end{aligned}$$

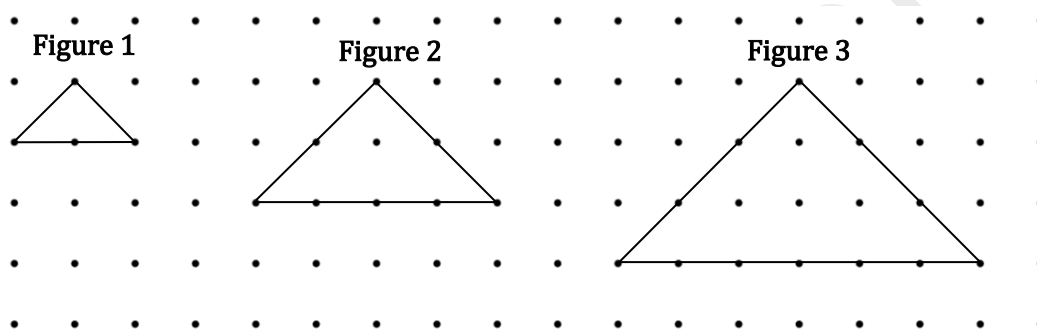


Total: 11 marks

8. An answer sheet is provided for this question.

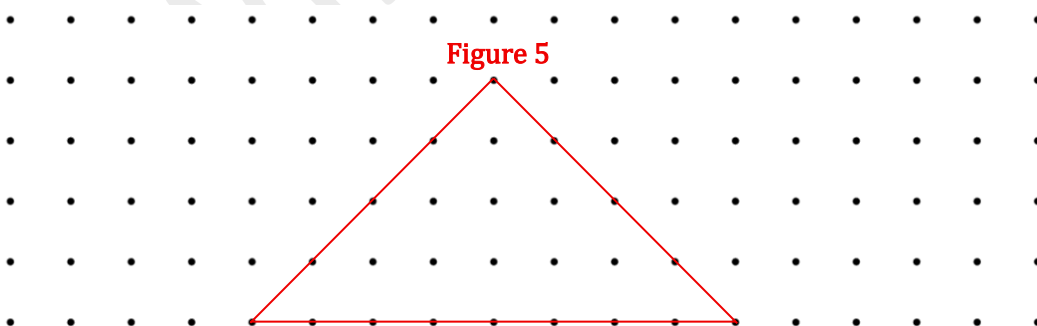
The diagram below shows the first three figures in a sequence of figures. Each figure is an isosceles triangle made of a rubber band stretched around pins on a geo-board.

The pins are arranged in rows and columns, one unit apart.



(a) On the answer sheet provided, draw the fourth figure (Figure 4) in the sequence.

[2]



(b) Study the pattern in the table below, and on your answer sheet, complete the rows numbered (i), (ii), (iii) and (iv). The breaks in the columns indicate that the rows do not follow one after the other.

	Figure	Area of Triangle	No. of Pins on Base	
	1	1	$2 \times 1 + 1 = 3$	
	2	4	$2 \times 2 + 1 = 5$	
	3	9	$2 \times 3 + 1 = 7$	
(i)	4	16	$(2 \times 4) + 1 = 9$	[2]
(ii)	10	100	$(2 \times 10) + 1 = 21$	[2]
(iii)	20	400	$(2 \times 20) + 1 = 41$	[2]
(iv)	n	n^2	$2n + 1$	[2]

Consider the n th figure.

Area of triangle = n^2

No. of Pins on Base = $2n + 1$

When $n = 4$,

Area of triangle = 4^2

= 16

When $n = 4$,

No. of Pins on Base = $2(4) + 1$

$$= 8 + 1$$

$$= 9$$

If the area of triangle = 100,

Then,

$$n^2 = 100$$

$$n = \sqrt{100}$$

$$n = 10$$

When $n = 10$,

No. of Pins on Base = $2(10) + 1$

$$= 20 + 1$$

$$= 21$$

When $n = 20$,

Area of triangle = 20^2

$$= 400$$

When $n = 20$,

No. of Pins on Base = $2(20) + 1$

$$= 40 + 1$$

$$= 41$$

Total: 10 marks

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SECTION II

There are **THREE** questions in this section.

Answer **TWO** questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) (i) Solve the pair of simultaneous equations:

$$y = 8 - x$$

$$2x^2 + xy = -16$$

[5]

$$y = 8 - x$$

...Equation 1

$$2x^2 + xy = -16$$

...Equation 2

Substituting Equation 2 into Equation 1 gives,

$$2x^2 + x(8 - x) = -16$$

$$2x^2 + 8x - x^2 = -16$$

$$x^2 + 8x + 16 = 0$$

$$x^2 + 4x + 4x + 16 = 0$$

$$x(x + 4) + 4(x + 4) = 0$$

$$(x + 4)(x + 4) = 0$$

We have

$$x + 4 = 0$$

$$x = -4$$

Substituting $x = -4$ into Equation 1 gives,

$$y = 8 - (-4)$$

$$y = 8 + 4$$

$$y = 12$$

$$\therefore x = -4 \text{ and } y = 12$$

- (ii) State, giving the reason for your answer, whether the line $y = 8 - x$ is tangent to the curve $2x^2 + xy = -16$. [2]

When $y = 8 - x$ and $2x^2 + xy = -16$ were solve simultaneously, only one solution was obtained. Since there is only one solution, the line ($y = 8 - x$) touches the curve ($2x^2 + xy = -16$) at exactly one point.

Therefore, the line ($y = 8 - x$) is a tangent to the curve ($2x^2 + xy = -16$).

When $x = -4$, $y = 12$ so the point of contact is $(-4, 12)$.

(b) An answer sheet is provided for this question.

A florist makes bouquets of flowers, each consisting of x roses and y orchids.

For each bouquet, she applies the following constraints:

- the number of orchids must be at least half the number of roses
- there must be at least 2 roses
- there must be no more than 12 flowers

(i) Write THREE inequalities for the constraints given. [3]

Constraint: The number of orchids must be at least half the number of roses.

Inequality: $y \geq \frac{1}{2}x$

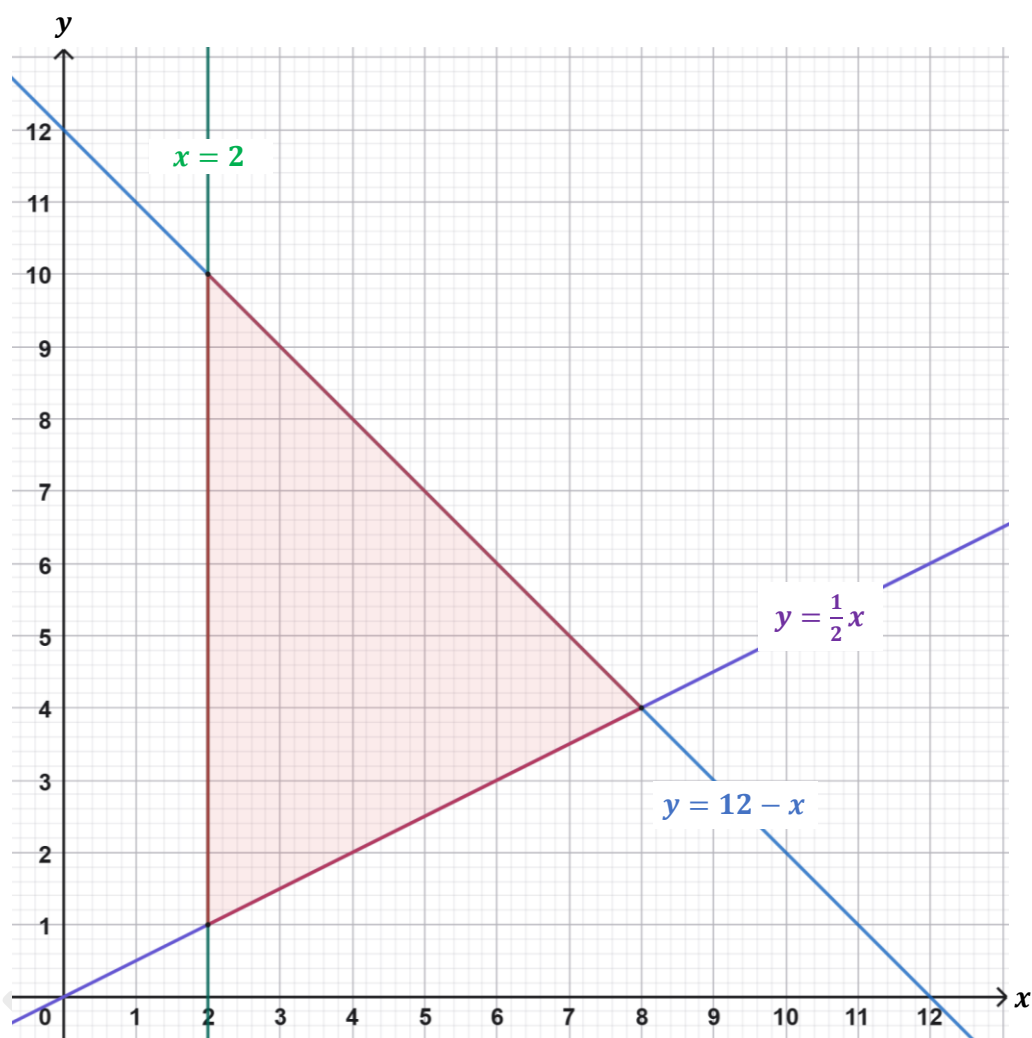
Constraint: There must be at least 2 roses.

Inequality: $x \geq 2$

Constraint: There must be no more than 12 flowers in the bouquet.

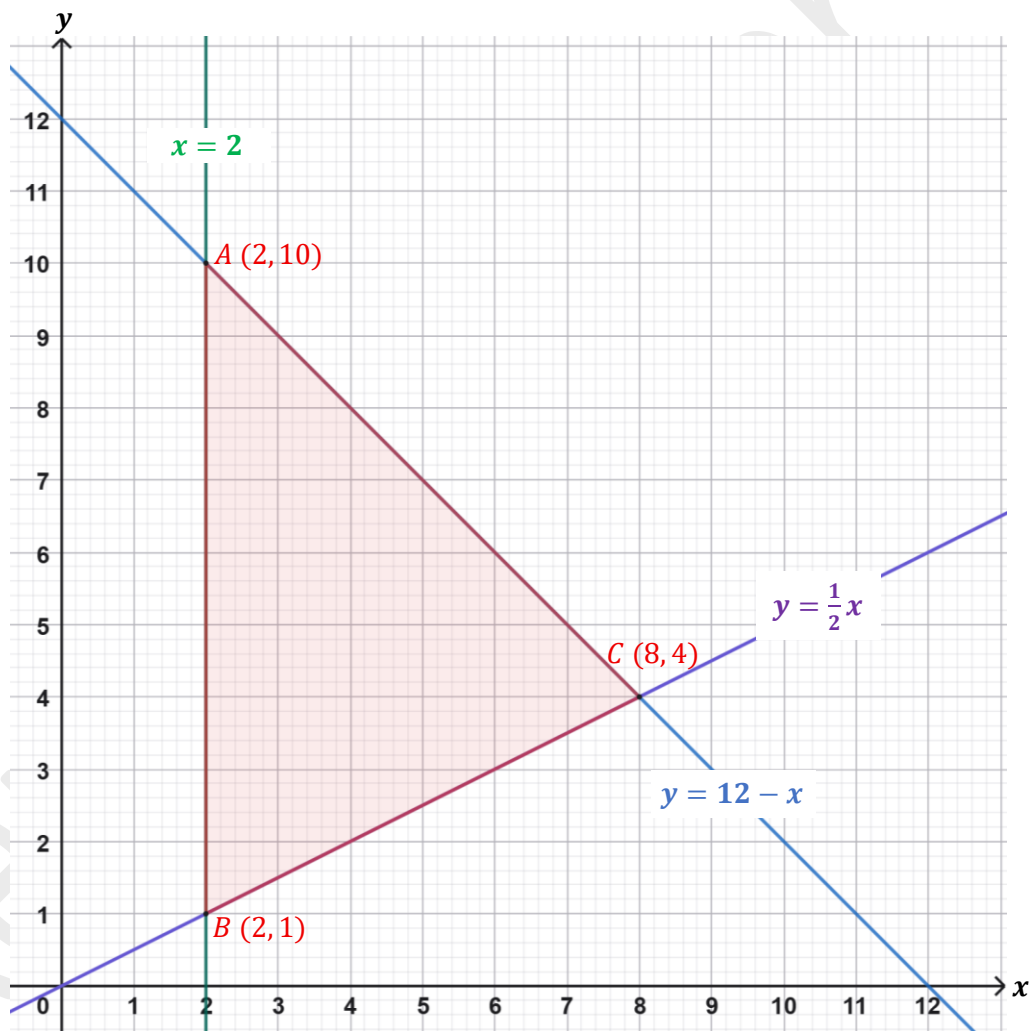
Inequality: $x + y \leq 12$

- (ii) On the answer sheet provided, shade the region of the graph which represents the solution set for the inequalities in (b)(i). [1]



- (iii) State the coordinates of the points which represent the vertices of the region showing the solution set. [1]

The vertices of the region are: $A(2, 10)$, $B(2, 1)$ and $C(8, 4)$



- (iv) The florist sells a bouquet of flowers to make a profit of \$3 on each rose and \$4 on each orchid. Determine the MAXIMUM possible profit on the sale of a bouquet. [3]

A profit of \$3 is made on each rose and \$4 on each orchid.

Let P be the profit.

Then, $P = 3x + 4y$.

We test the three points at the three vertices of the feasible region.

For the point $A(2, 10)$, $x = 2$ and $y = 10$.

$$P = 3(2) + 4(10)$$

$$P = 6 + 40$$

$$P = 46$$

For the point $B(2, 1)$, $x = 2$ and $y = 1$.

$$P = 3(2) + 4(1)$$

$$P = 6 + 4$$

$$P = 10$$

For the point $C(8, 4)$, $x = 8$ and $y = 4$.

$$P = 3(8) + 4(4)$$

$$P = 24 + 16$$

$$P = 40$$

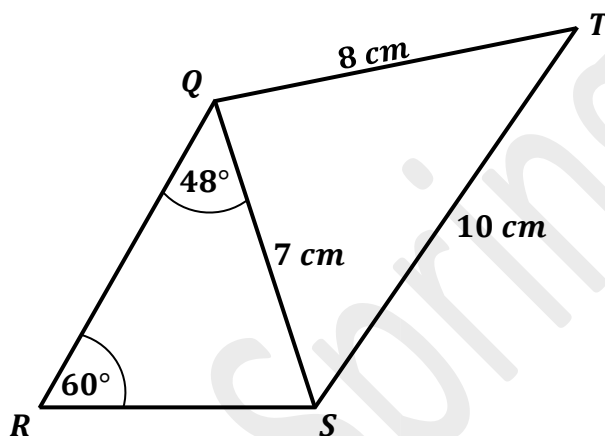
$\therefore$ The maximum profit occurs at point $A(2, 10)$ where the profit is \$46,
the number of roses is 2 and the number of orchids is 10.

Total: 15 marks

MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) The diagram below, **not drawn to scale**, shows a quadrilateral $QRST$ in which

$QS = 7 \text{ cm}$, $ST = 10 \text{ cm}$, $QT = 8 \text{ cm}$, $\angle SRQ = 60^\circ$ and $\angle RQS = 48^\circ$.



Calculate

(i) the length of RS

[3]

Consider triangle QRS . Using the sine rule,

$$\frac{QS}{\sin \hat{R}} = \frac{RS}{\sin \hat{Q}}$$

$$\frac{7}{\sin 60^\circ} = \frac{RS}{\sin 48^\circ}$$

$$RS \sin 60^\circ = 7 \sin 48^\circ$$

$$RS = \frac{7 \sin 48^\circ}{\sin 60^\circ}$$

$$RS = 6.01 \quad (\text{to 2 decimal places})$$

- (ii) the measure of $\angle QTS$

Consider the triangle QTS . Using the cosine rule,

$$QS^2 = QT^2 + TS^2 - 2(QT)(TS) \cos \hat{T}$$

$$(7)^2 = (8)^2 + (10)^2 - 2(8)(10) \cos \hat{T}$$

$$49 = 64 + 100 - 160 \cos \hat{T}$$

$$49 = 164 - 160 \cos \hat{T}$$

$$\cos \hat{T} = \frac{164-49}{160}$$

$$\cos \hat{T} = 0.71875$$

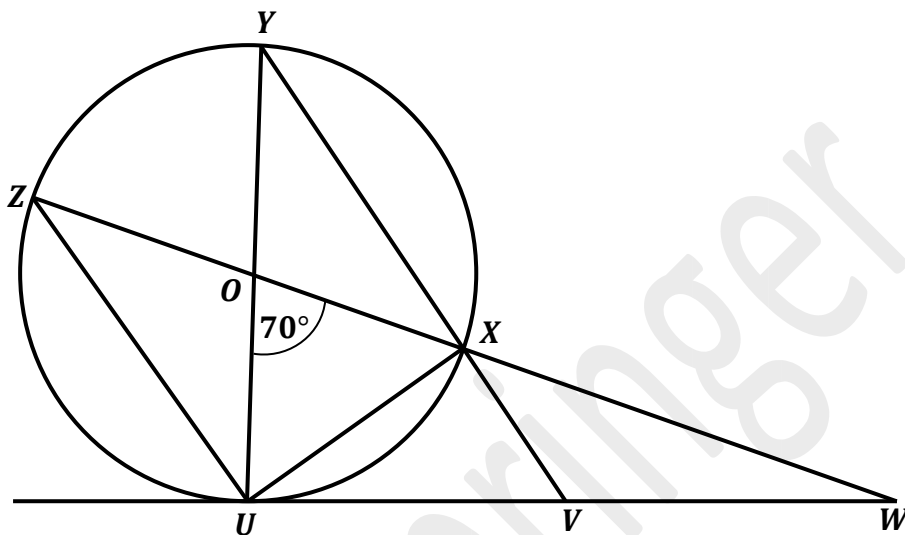
$$\hat{T} = \cos^{-1}(0.71875)$$

$$\hat{T} = 44.0^\circ \quad (\text{to 1 decimal place})$$

$$\therefore \angle QTS = 44^\circ$$

(b) The diagram below, **not drawn to scale**, shows a circle, centre O . The line

UVW is a tangent to the circle, $ZOXW$ is a straight line and angle $UOX = 70^\circ$.



(i) Calculate, showing working where necessary, the measure of angle

(a) $\angle OUZ$

[2]

The sum of angles on a straight line add up to 180° .

$$\begin{aligned}\angle OUZ &= 180^\circ - 70^\circ \\ &= 110^\circ\end{aligned}$$

Since OZ and OU are radii of the same circle, $OZ = OU$.

Therefore, $\triangle ZOU$ is an isosceles triangle. The base angles of an isosceles triangle are equal and the sum of the three interior angles of a triangle add up to 180° .

$$\begin{aligned}\therefore \widehat{O\hat{U}Z} &= \frac{180^\circ - 110^\circ}{2} \\ &= 35^\circ\end{aligned}$$

(b) UVY

[3]

The angle at the centre of a circle is twice the angle at the circumference of the circle from the same chord.

$$\begin{aligned}\widehat{U\hat{V}Y} &= \frac{70^\circ}{2} \\ &= 35^\circ\end{aligned}$$

The angle made by the tangent UW to a circle and a radius OU is a right angle.

$$\therefore \widehat{O\hat{U}V} = 90^\circ$$

The sum of angles in a triangle add up to 180° .

$$\begin{aligned}\widehat{U\hat{V}Y} &= 180^\circ - (90^\circ + 35^\circ) \\ &= 180^\circ - 125^\circ \\ &= 55^\circ\end{aligned}$$

(c) UWO

The sum of angles in a triangle add up to 180° .

$$\widehat{WOU} + \widehat{OWU} + \widehat{UWO} = 180^\circ$$

$$70^\circ + 90^\circ + \widehat{UWO} = 180^\circ$$

$$\widehat{UWO} = 180^\circ - (70^\circ + 90^\circ)$$

$$\widehat{UWO} = 180^\circ - 160^\circ$$

$$\widehat{UWO} = 20^\circ$$

(ii) Name the triangle in the diagram which is congruent to triangle

(a) ZOU

[1]

Since OZ and OY are radii of the same circle, $OZ = OY$.

Since OU and OX are radii of the same circle, $OU = OX$.

Since vertically opposite angles are equal, then $\widehat{ZOU} = \widehat{YOX}$.

Hence, $\triangle ZOU$ is congruent to $\triangle YOX$.

(b) YXU

Consider ΔYXU and ΔZUX .

Since YU and ZX are both diameter of the same circle, then $YU = ZX$.

Since the angle in a semicircle is equal to 90° , then $\angle YXU = \angle ZUX = 90^\circ$.

UX is a common side to both triangles.

So, both right angled triangles, YXU and ZUX have the same hypotenuse and share a common side.

Hence, ΔYXU is congruent to ΔZUX .

Total: 15 marks

VECTORS AND MATRICES

11. (a) The points A , B and C have position vectors $\overrightarrow{OA} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and

$\overrightarrow{OC} = \begin{pmatrix} 12 \\ -2 \end{pmatrix}$ respectively.

(i) Express in the form $\begin{pmatrix} x \\ y \end{pmatrix}$ the vector

(a) $\overrightarrow{BA}$

[2]

$$\begin{aligned}
 \overrightarrow{BA} &= \overrightarrow{BO} + \overrightarrow{OA} \\
 &= -\overrightarrow{OB} + \overrightarrow{OA} \\
 &= -\begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} \\
 &= \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} \\
 &= \begin{pmatrix} -3 + 6 \\ -4 + 2 \end{pmatrix} \\
 &= \begin{pmatrix} 3 \\ -2 \end{pmatrix}
 \end{aligned}$$

which is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where $x = 3$ and $y = -2$.

(b) $\overrightarrow{BC}$

$$\begin{aligned}
 \overrightarrow{BC} &= \overrightarrow{BO} + \overrightarrow{OC} \\
 &= -\overrightarrow{OB} + \overrightarrow{OC} \\
 &= -\begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 12 \\ -2 \end{pmatrix} \\
 &= \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \begin{pmatrix} 12 \\ -2 \end{pmatrix} \\
 &= \begin{pmatrix} -3 + 12 \\ -4 - 2 \end{pmatrix} \\
 &= \begin{pmatrix} 9 \\ -6 \end{pmatrix}
 \end{aligned}$$

which is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where $x = 9$ and $y = -6$.

(ii) State ONE geometrical relationship between BA and BC . [1]

$$\begin{aligned}
 \overrightarrow{BA} &= \begin{pmatrix} 3 \\ -4 \end{pmatrix} \\
 \overrightarrow{BC} &= \begin{pmatrix} 9 \\ -6 \end{pmatrix} \\
 &= 3 \begin{pmatrix} 3 \\ -4 \end{pmatrix}
 \end{aligned}$$

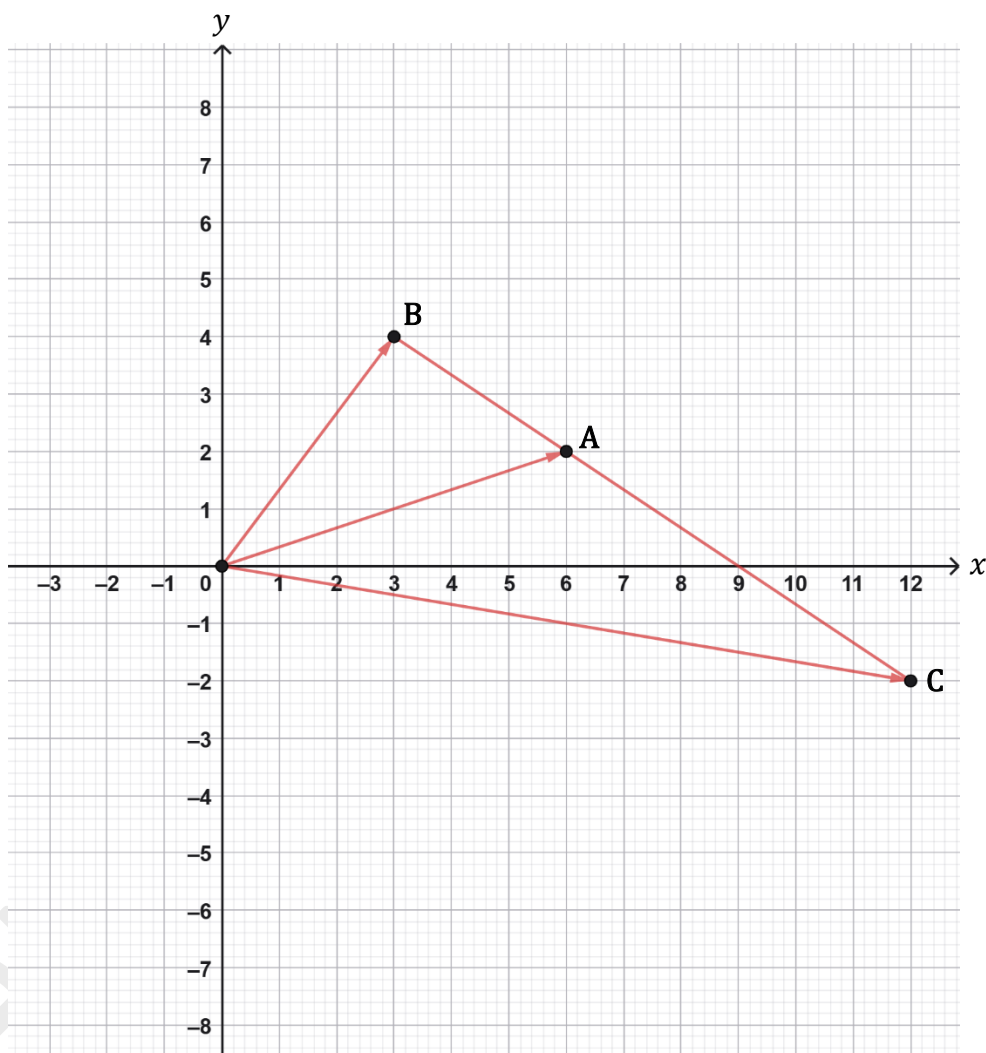
$$\text{So, } |\overrightarrow{BC}| = 3|\overrightarrow{BA}|.$$

Since $\overrightarrow{BA}$ can be represented as a scalar multiple of $\overrightarrow{BC}$, and B is a

common point on both vectors, then $\overrightarrow{BA}$ and $\overrightarrow{BC}$ are parallel.

Therefore, A , B and C are collinear and $\overrightarrow{BC}$ is three times the length of $\overrightarrow{BA}$.

- (iii) Draw a sketch to show the relative positions of A , B and C . [2]



(b)(i) Calculate the values of a and b such that $\begin{pmatrix} a & -4 \\ 1 & b \end{pmatrix} \begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. [3]

$$\begin{pmatrix} a & -4 \\ 1 & b \end{pmatrix} \begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} (a \times 2) + (-4 \times 1) & (a \times -4) + (-4 \times -3) \\ (1 \times 2) + (b \times 1) & (1 \times -4) + (b \times -3) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2a - 4 & -4a + 12 \\ 2 + b & -4 - 3b \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

Comparing the equivalent matrices and equating corresponding entries, we get:

$$2a - 4 = 2$$

and

$$2 + b = 0$$

$$2a = 4 + 2$$

$$b = -2$$

$$2a = 6$$

$$a = \frac{6}{2}$$

$$a = 3$$

$$\therefore a = 3 \text{ and } b = -2.$$

(ii) Hence, or otherwise, write down the inverse of $\begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix}$.

$$\det(A) = ad - bc$$

$$= (2)(-3) - (-4)(1)$$

$$= -6 - (-4)$$

$$= -6 + 4$$

$$= -2$$

$$\text{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 4 \\ -1 & 2 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{\det} \times \text{adj}(A)$$

$$= -\frac{1}{2} \times \begin{pmatrix} -3 & 4 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{-3}{-2} & \frac{4}{-2} \\ \frac{-1}{-2} & \frac{2}{-2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{2} & -2 \\ \frac{1}{2} & -1 \end{pmatrix}$$

(iii) Use the inverse of $\begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix}$ to solve for x and y in the matrix equation

$$\begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ 7 \end{pmatrix}. \quad [3]$$

$$\begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & -4 \\ 1 & -3 \end{pmatrix}^{-1} \begin{pmatrix} 12 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -3 & 4 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 12 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} (-3 \times 12) + (4 \times 7) \\ (-1 \times 12) + (2 \times 7) \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -36 + 28 \\ -12 + 14 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -8 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$\therefore x = 4 \text{ and } y = -1.$$

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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